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/* ========================================================================
 * Project: Captain's 5-Phase CNC Motor Controller (Pocket CNC 5x5)
 * MCU: PIC16F73 | Oscillator: 20 MHz External Crystal
 * Compiler: mikroC PRO for PIC
 * Logic: Inverted for PC817 Optocoupler (1k Pull-up on Collector)
 * ======================================================================== */

// --- PIN CONFIGURATION ---
// CNC Input Pins (via PC817 - Inverted Logic)
#define STEP_IN     RA0_bit     // CNC  (0 = Active Edge, 1 = Idle)
#define DIR_IN      RA1_bit     // CNC  (0 = Reverse/Clockwise, 1 = Forward)
#define ENABLE_IN   RA2_bit     // CNC  (Low to Enable)

// Manual Control Buttons (Direct - Normal Logic)
#define BTN_FWD     RA3_bit     //  (Active High)
#define BTN_RWD     RA4_bit     // (Active High)
#define MODE_FH     RA5_bit     // 0 = Full-Step (0.72°), 1 = Half-Step (0.36°)

// Driver Output Pins (To IR2103 Gate Drivers)
// PORTB: 5 Pins for Normal Outputs (A, B, C, D, E)
// PORTC: 5 Pins for Inverted/Complementary Outputs (~A, ~B, ~C, ~D, ~E)
#define OUT_B       PORTB
#define OUT_C       PORTC

// --- 5-PHASE PENTAGON DRIVE LUT (Look-Up Tables) ---
// Full-Step Sequence (10 Steps - 4 Phase ON at a time, 0.72 deg)
const unsigned char Full_Step_B[10] = {
    0b00011001, 0b00011100, 0b00001100, 0b00001110, 0b00000110,
    0b00000111, 0b00000011, 0b00010011, 0b00010001, 0b00011001
};
const unsigned char Full_Step_C[10] = {
    0b00000110, 0b00000011, 0b00010011, 0b00010001, 0b00011001,
    0b00011100, 0b00001100, 0b00001110, 0b00000110, 0b00000111
};

// Half-Step Sequence (20 Steps - Alternating 4 and 5 Phase ON, 0.36 deg)
const unsigned char Half_Step_B[20] = {
    0b00011001, 0b00011011, 0b00011100, 0b00011101, 0b00001100,
    0b00001110, 0b00001111, 0b00000110, 0b00000111, 0b00000111,
    0b00000011, 0b00010011, 0b00010011, 0b00010001, 0b00011001,
    0b00011001, 0b00011011, 0b00011100, 0b00011101, 0b00001100
};
const unsigned char Half_Step_C[20] = {
    0b00000110, 0b00000111, 0b00000011, 0b00010011, 0b00010011,
    0b00010001, 0b00011001, 0b00011001, 0b00011011, 0b00011100,
    0b00011101, 0b00001100, 0b00001110, 0b00001111, 0b00000110,
    0b00000111, 0b00000111, 0b00000011, 0b00010011, 0b00010011
};

// --- MOTOR HELPER FUNCTIONS ---
void Motor_Rest() {
    OUT_B = 0x00;
    OUT_C = 0x00;
}

void Output_Full_Step(int step_index) {
    OUT_B = Full_Step_B[step_index];
    OUT_C = Full_Step_C[step_index];
}

void Output_Half_Step(int step_index) {
    OUT_B = Half_Step_B[step_index];
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 63:     OUT_C = Half_Step_C[step_index];
 64: }
 65:
 66: // --- MAIN FUNCTION ---
 67: void main() {
 68:     // Variable Declarations
 69:     int step_f = 0;
 70:     int step_h = 0;
 71:     char prev_step = 1; // PC817  (High)
 72:
 73:     // Microcontroller Registers Initialization
 74:     ADCON1 = 0x06;      // (PORTA)
 75:     CCP1CON = 0x00;     // CCP1  (PWM )
 76:
 77:     TRISA  = 0xFF;      // PORTA (CNC, Buttons & Mode)
 78:     TRISB  = 0x00;      // PORTB  (Normal Phase Outputs)
 79:     TRISC  = 0x00;      // PORTC  (Complementary Phase Outputs)
 80:
 81:     Motor_Rest();
 82:
 83:     // --- INFINITE LOOP ---
 84:     while(1) {
 85:
 86:
 87:         if(ENABLE_IN == 0) {
 88:
 89:             // ------------------------------------------------------
 90:             // ?. MANUAL BUTTON CONTROL (CNC )
 91:             // ------------------------------------------------------
 92:             if(BTN_FWD == 1 && BTN_RWD == 0) {
 93:                 if(MODE_FH == 0) {
 94:                     step_f++; if(step_f > 9) step_f = 0;
 95:                     Output_Full_Step(step_f);
 96:                 } else {
 97:                     step_h++; if(step_h > 19) step_h = 0;
 98:                     Output_Half_Step(step_h);
 99:                 }
100:                 Delay_ms(5);
101:             }
102:
103:             else if(BTN_RWD == 1 && BTN_FWD == 0) {
104:                 if(MODE_FH == 0) {
105:                     step_f--; if(step_f < 0) step_f = 9;
106:                     Output_Full_Step(step_f);
107:                 } else {
108:                     step_h--; if(step_h < 0) step_h = 19;
109:                     Output_Half_Step(step_h);
110:                 }
111:                 Delay_ms(5);
112:             }
113:
114:             // ------------------------------------------------------
115:             // ?. CNC AUTO CONTROL (OPTOCOUPLER INVERTED LOGIC)
116:             // ------------------------------------------------------
117:             else {
118:                 // STEP_IN == 1,  STEP_IN == 0
119:                 // PC817 0V (Low)
120:                 if(STEP_IN == 0 && prev_step == 1) {
121:
122:                     // DIR_IN :
123:                     // DIR_IN == 0  CNC  1 (High)
124:                     if(DIR_IN == 0) {
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125:                         if(MODE_FH == 0) {
126:                             step_f++; if(step_f > 9) step_f = 0;
127:                             Output_Full_Step(step_f);
128:                         } else {
129:                             step_h++; if(step_h > 19) step_h = 0;
130:                             Output_Half_Step(step_h);
131:                         }
132:                     }
133:
134:                     else {
135:                         if(MODE_FH == 0) {
136:                             step_f--; if(step_f < 0) step_f = 9;
137:                             Output_Full_Step(step_f);
138:                         } else {
139:                             step_h--; if(step_h < 0) step_h = 19;
140:                             Output_Half_Step(step_h);
141:                         }
142:                     }
143:                 }
144:                 prev_step = STEP_IN;
145:             }
146:
147:         } else {
148:             Motor_Rest();
149:         }
150:     }
151: }
```